

NFR Traceability Matrix

ID	Quantified requirement	Tactic in SAS	Test / tool	Target	Actual
	Performance : p95 of request at 100 concurrent users	- Redis caching - Database indexing -Asynchronous processing - Optimized JPA queries using JOINS - Stateless services layer	Apache JMeter Spring Boot Actuator Prometheus AWS CloudWatch	=< 2.0s (Error rate < 1%)	
	Scalability: 200% workload increase with ≤ 10% degradation	- Stateless services layer - Database read replicas	Apache JMeter Prometheus AWS CloudWatch	=< 10% degradation	
	Security: 100% authorization required	- Spring Security with JWT - RBAC enforcement - AWS Parameter Store - BCrypt password hashing	Spring Security JWT for stateless authentication AWS Parameter Store OWASP ZAP	100% authorized access	

	Security: MFA required for elevated privileges	- MFA enforcement for admin accounts			
	Availability over the 30 ≥ 99.9% uptime	- Spring Boot Actuator health endpoints - Auto-restart on failure - AWS Application Load Balancer health checks	AWS CloudWatch, UptimeRobot	≥ 99.9% uptime	
	Availability: Minimum 2hr recovery	- Automated backups -Automated Failover replication	AWS Backup, PostgreSQL Multi-AZ	Downtime =< 2 hours	
	Availability: Minimum 1% data loss	- ACID transactions - Foreign key constraints	JUnit integration tests PostgreSQL transaction logs AWS Backup	Data loss < 0.01%)	
	Maintainabili ty: 80% test coverage	- JUnit 5 and Mockito for unit tests - Jest for frontend tests - JaCoco for coverage reporting	JUnit 5, Mockito, Jest, JaCoCo	≥ 80% coverage	

		- Automated CI/CD execution			
	Maintainability: Isolated adapter changes	- Layered architecture - Adapter pattern for third party integrations - ArchUnit for architectural enforcement	ArchUnit, SonarQube	Zero architectural violation warnings	